

Trigonometric Identities — Cheat Sheet

Basic / Definitions

Reciprocals: $\sec x = \frac{1}{\cos x}$, $\csc x = \frac{1}{\sin x}$, $\cot x = \frac{\cos x}{\sin x}$.

Quotients: $\tan x = \frac{\sin x}{\cos x}$, $\cot x = \frac{1}{\tan x}$.

Pythagorean: $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \csc^2 x$.

Negative angle: $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$, $\tan(-x) = -\tan x$.

Periodicity: $\sin(x + 2\pi) = \sin x$, $\cos(x + 2\pi) = \cos x$, $\tan(x + \pi) = \tan x$.

Angle Addition / Subtraction

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y,$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y,$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}.$$

Double / Triple Angle

$$\sin(2x) = 2 \sin x \cos x,$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x,$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x} \quad (\tan x \neq \pm 1),$$

$$\sin(3x) = 3 \sin x - 4 \sin^3 x, \quad \cos(3x) = 4 \cos^3 x - 3 \cos x.$$

Half-Angle / Power-Reducing

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \quad \cos^2 x = \frac{1 + \cos(2x)}{2},$$

$$\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}, \quad \cos^2 \frac{x}{2} = \frac{1 + \cos x}{2},$$

$$\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}.$$

Product-to-Sum & Sum-to-Product

$$\sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)],$$

$$\cos x \cos y = \frac{1}{2} [\cos(x+y) + \cos(x-y)],$$

$$\sin x \sin y = \frac{1}{2} [\cos(x-y) - \cos(x+y)],$$

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}, \quad \sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2},$$

$$\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}, \quad \cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}.$$

Useful for Integration

$$\int \sin^2 x \, dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C, \quad \int \cos^2 x \, dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C,$$

$$\int \sin x \cos x \, dx = -\frac{\cos^2 x}{2} + C = \frac{\sin^2 x}{2} + C,$$

$$\sin^2 x \cos^2 x = \frac{1}{4} \sin^2(2x), \quad \int \sin^2 x \cos^2 x \, dx = \frac{x}{8} - \frac{\sin(4x)}{32} + C.$$

Complex (Euler) Forms

$$e^{ix} = \cos x + i \sin x, \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

Parity tip: $\cos$ is even, $\sin$ is odd. On $[-L, L]$, integrals of odd functions vanish; even functions double: $\int_{-L}^L \text{even} = 2 \int_0^L$.